



Breach Series

Part 1: Breach Point

Project Breach Point

First part of the breach series under the Penetration Testing track simulates the tactics, techniques, and procedures (TTPs) of a threat actor. This project challenges you to apply the penetration testing lifecycle from information gathering, exploitation, to privilege escalation and data exfiltration. The goal is to automate parts of the attack using .sh scripts to simulate the attacker activity, ensuring repeatability and traceability.

Project Structure:

Scope 1

1. Getting the User Input
 - 1.1. Get from the user a network to scan.
 - 1.2. Get from the user a name for the output directory, if doesn't exist create one.
 - 1.3. Allow the user to choose 'Basic' or 'Full'.
 - 1.4. Basic: scans the network for TCP and UDP, including the service version and weak passwords.
 - 1.5. Full: include Nmap Scripting Engine (NSE), weak passwords, and vulnerability analysis.
 - 1.6. Display potential vulnerabilities via NSE and Searchsploit.
2. Present the attack paths and let the user choose
 - 2.1. Look for weak login credentials (refer to part 3)
 - 2.2. Create a .rc file to use Metasploit to automate the use of exploits / suggerter / handler (refer to part 4)
 - 2.3. Payload Generation (refer to part 5)
 - 2.4. Data Exfiltration (refer to part 6)
3. Weak Credentials
 - 3.1. Look for weak passwords used in the network for login services.
 - 3.2. Have a built-in password.lst to check for weak passwords.
 - 3.3. Allow the user to provide their own password list.
 - 3.4. Login services to check include: SSH, RDP, FTP, and SMB.
4. Automate with resource file
 - 4.1. Allow the user to choose which .rc file to generate to do the following:
 - 4.1.1. Exploit ssh login with auxiliary/scanner/ssh/ssh_login module or let the user select another module
 - 4.1.2. Create a handler
 - 4.1.3. Use the suggerter
 - 4.2. Allow the user to choose if they want to execute the .rc file
5. Generate a payload
 - 5.1. Get from the user which directory to store the payload
 - 5.2. Get from the user the payload name, LHOST, LPORT, format, output name and any other features if applicable.
6. Data Exfiltration
 - 6.1. Get the user to specify the Operating System: Linux or Windows
 - 6.2. Depending on the OS specified, generate the commands for the user to look for files containing these words: [password, .docx, .xlsx.] Use the command find or dir for Linux and Windows respectively.
 - 6.3. Generate the commands to compress into a .zip file
 - 6.4. Generate the commands to encode into base64
 - 6.5. Generate the command to scp to the attacker machine
7. Log Results
 - 7.1. During each stage, display the stage in the terminal.

7.2. At the end, summarise what was done during that session.

7.3. Allow the user to search inside the results or exit.

General Tools available: nmap, NSEs, searchsploit, Metasploit, msfvenom, msfconsole, hydra, case scripting, find, dir, certutil etc.

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Project Deliverables:

- Record the execution of the .sh to show how the script work into an .mp4 format.
- Use comments in your code to explain what you did.
- If you are using code from the internet, add credit and links\
- Put the script (.sh) and pdf report into a Zip File.
- Upload the .zip and recording to drive
- File Name: UNIT.STUDENT NUMBER (e.g. CCK1_240101.s5.zip, CCK1_240101.s5.sh, CCK1_240101.s5.pdf etc.)
- Failure to do so will be taken as no submission and a 0 grade for the project
- Upload the project to your individual google drive folder
- Last Submission: 23:59 of the project submission deadline. Folder will be denied access thereafter.